

Supporting Information

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SI Materials and Methods

Dynamic Model of Three-Member Consortia. The three-member kinetic model was simulated using the Matlab environment. The model was initially seeded with 10^7 cells of each strain. Cooperativity coefficients were taken from pairwise cooperativity coefficients (c_{12} , c_{21}) provided in [Dataset S2](#). Predicted fold growth was calculated by dividing time step $t = 5,500$ population values by the initial seeding value. The carrying capacity was set to 10^9 cells in all simulations. The β value was set to 1. Comparison between the predicted fold growth and observed fold growth is shown in Fig. S34. In general, we find high predictive power in a significant fraction of the consortia (Pearson coefficient $r = 0.51$, P value = 5.2×10^{-25}). Differences between the predicted and observed fold growth highlights possible positive and negative epistatic interactions (Fig. S3B).

Dynamic Model of 14-Member Consortia. The 14-member kinetic model was simulated using the pairwise cooperativity coefficients (c_{12} , c_{21}) provided in [Dataset S2](#). The population is seeded with equal amount of all 14 auxotrophs totaling 10^7 cells. The carrying capacity was set to 10^9 cells in all simulations. The β value was set to 1. The resulting steady-state population abundance was determined as the percentage of the total population dominated by each auxotroph type (Fig. S44). For each 13-member dropout simulations, each dropout auxotroph was set to an initial seeding

value of 0, and all other aspects of the simulation were unchanged. The difference in population abundance for each auxotroph in the 13-member groups compared with the 14-member group is shown in Fig. S4B.

Biosynthesis Predictions from the Joint Genome Institute Integrated Microbial Genomes Database. Amino acids biosynthetic predictions are pulled from the Integrated Microbial Genomes (IMG) database “phenotype function” system (1). This bioinformatic analysis pipeline first predicts the enzymatic capabilities of a given organism. Next, logical testing is used for prototrophy calls to ensure that for each reaction step in amino acid biosynthesis from a common precursor (e.g., pyruvate for leucine biosynthesis), there is a predicted enzyme that can catalyze the reaction. Similarly for auxotroph calls, an organism must be lacking enzymes catalyzing at least one of the steps in the biosynthesis pathway. Attempts have been made to take into account alternate biosynthesis pathways (e.g., methionine synthesis via homocysteine or methanethiol intermediates), although undoubtedly not every exception will be captured and these computational approaches will not have 100% accuracy. Nonetheless, this is a powerful tool to make general trend observations. For details on prediction rule, please see the IMG website.

1. Markowitz VM, et al. (2012) IMG: The Integrated Microbial Genomes database and comparative analysis system. *Nucleic Acids Res* 40(Database issue):D115–D122.

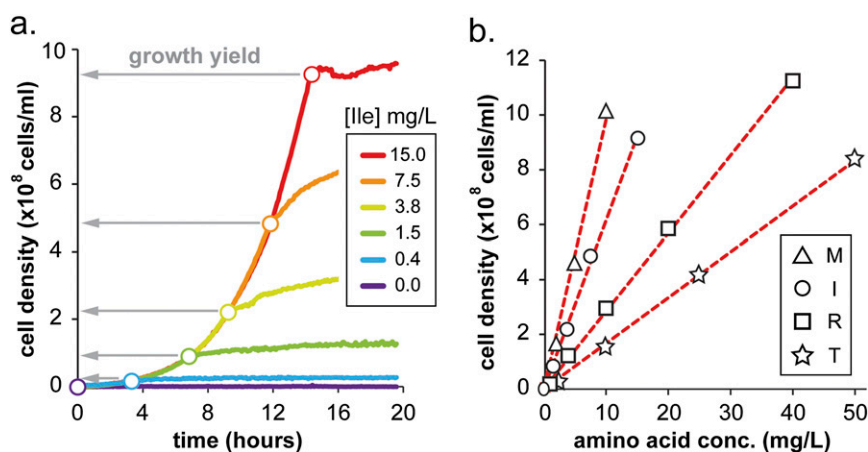


Fig. S1. Calculation of amino acid utilization during growth. (A) Example growth curves of *Escherichia coli* isoleucine auxotroph ($\Delta ilvA$) in M9-glucose supplemented with varying initial concentrations of free L-isoleucine. The open circles denote the end of exponential phase resulting from supplemented amino acid exhaustion and marks final growth yield. (B) Linear relationship between amounts of supplemented amino acid and the observed growth yield for auxotrophs M, I, R, and T. Beyond the highest indicated amino acid concentrations, the yield response saturated due to other metabolites becoming growth limiting. The slope of the linear relationship between amino acid concentration and cell density is used to calculate the amino acids required per auxotrophic cell in M9 media presented in [Dataset S1](#).

Fig. S2. Growth yield of 91 pairwise cocultures shows lack of relationship between relative population abundance and syntrophic growth. (A) Coculture growth of each of 91 pairwise two-member consortia. Left ordinate axis (in blue) shows whole coculture fold growth after 84 h for each of 91 consortia (data in blue dots). Right ordinate axis (in orange) shows fraction of strain 1 in population for each consortia (data in orange dots). Consortia are ranked from lowest to highest based fraction of population dominated by strain 1. The pairwise coculture index number indicated on the x axis refers to the coculture ID presented in [Dataset S2](#). (B) Calculated average coculture growth for each auxotroph paired with all 13 other possible auxotrophs. The left ordinate axis (in black) shows whole coculture fold growth after 84 h for each auxotroph, which is designated strain 1. The color stacked bars indicates average fold growth contributed by strain 1 (in black) and strain 2 (in gray). The right ordinate axis (in orange) shows average fraction of strain 1 in population for each auxotroph group (data in orange range).

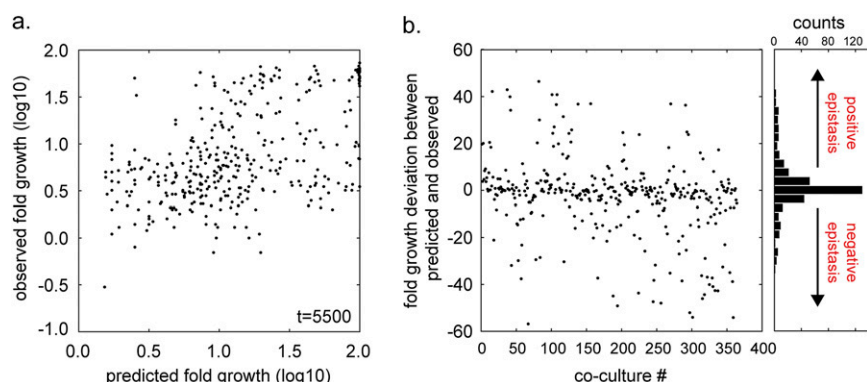


Fig. S3. Evaluation of dynamic growth model of three-member consortia. (A) Scatter plot comparing predicted fold growth of three-member consortia (in $\log_{10}$ units) versus observed fold growth of three-member consortia (in $\log_{10}$ units). Each black dot corresponds to each of the 364 possible unique consortia. (B) Scatter plot showing the difference between predicted and expected fold growth for all 364 consortia. Histogram on the *Right* shows the distribution across all consortia indicating general agreement between predictions and observations. Positive values indicate potential presence of positive epistasis, whereas negative values indicate potential negative epistasis.

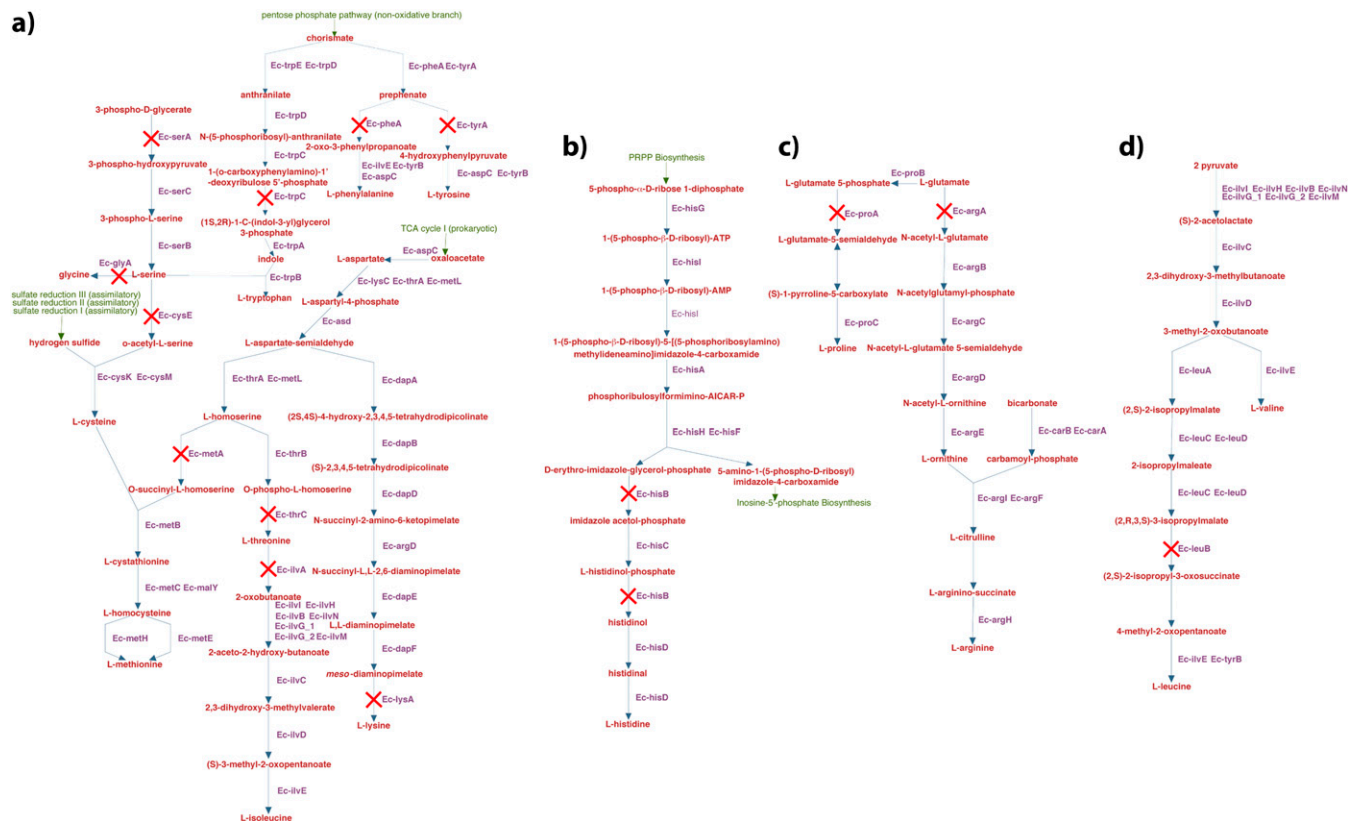


Fig. S5. Metabolic network diagrams. Biosynthesis reactions disrupted to generate auxotrophies are indicated with red crosses. Figures modified from Metacyc (1). Biosynthesis pathways for (A) C, F, G, I, K, M, S, T, W, Y; (B) H; (C) P, R; (D) L.

1. Caspi R, et al. (2014) The MetaCyc database of metabolic pathways and enzymes and the BioCyc collection of Pathway/Genome Databases. *Nucleic Acids Res* 42(Database issue):D459–D471.

Dataset S1. List of amino acid auxotrophs, the amount of supplemented amino acids needed to reproduce one cell, and the metabolic cost to biosynthesize each molecule of amino acid in units of phosphate bonds used

[Dataset S1](#)

Amino acid requirement per cell is calculated using the slope from Fig. S1B as follows: $(1/(\text{slope} \times \text{amino acid F.W. in g/mol} \times 10^6)) \times 6.022 \times 10^{23}$.

Dataset S2. Fold growth values of two-member and three-member consortia

[Dataset S2](#)

Growth values for two-member consortia are given for each strain as determined by quantitative PCR (*Materials and Methods*) and the determined cooperative coefficients are also shown. Total coculture growth values for each three-member consortia are given along with the three corresponding two-member subsets based on the pairwise data. Average cell concentrations for three or four replicates of the 14- and 13-member cultures are provided for the 7 d of passaging. Relative abundance for the 50-d passaging of the 14-member consortia is shown for the two replicates displayed in Fig. 4C.

Dataset S3. Detailed results from the 3- and 14-member dynamic modeling

[Dataset S3](#)

Fit between the three-member dynamic model and experimental results are listed for removal of certain groups of data points as defined by the zones in Fig. 3B. Sensitivity analysis for the initial cell concentration for the 14-member dynamic growth simulation is included. Sensitivity analysis 1 shows the result of stochastically seeding all strains with a normal distribution centered at $1e7$ cells and SDs ranging from 10% to 100% of the mean as indicated. Similarly, sensitivity analysis 2 shows the results of introducing stochasticity (seeding density pulled from a normal distribution centered at $1e7$ cells and SD of 50% of the mean) to each strain in the mixture independently. This is to assess whether any single strain has more of an effect than the others. The stability of the model demonstrated by these analyses suggests that any divergence with the experimental results is unlikely to be due to path dependence effects of the model.

Dataset S4. List of primers used in this study

[Dataset S4](#)